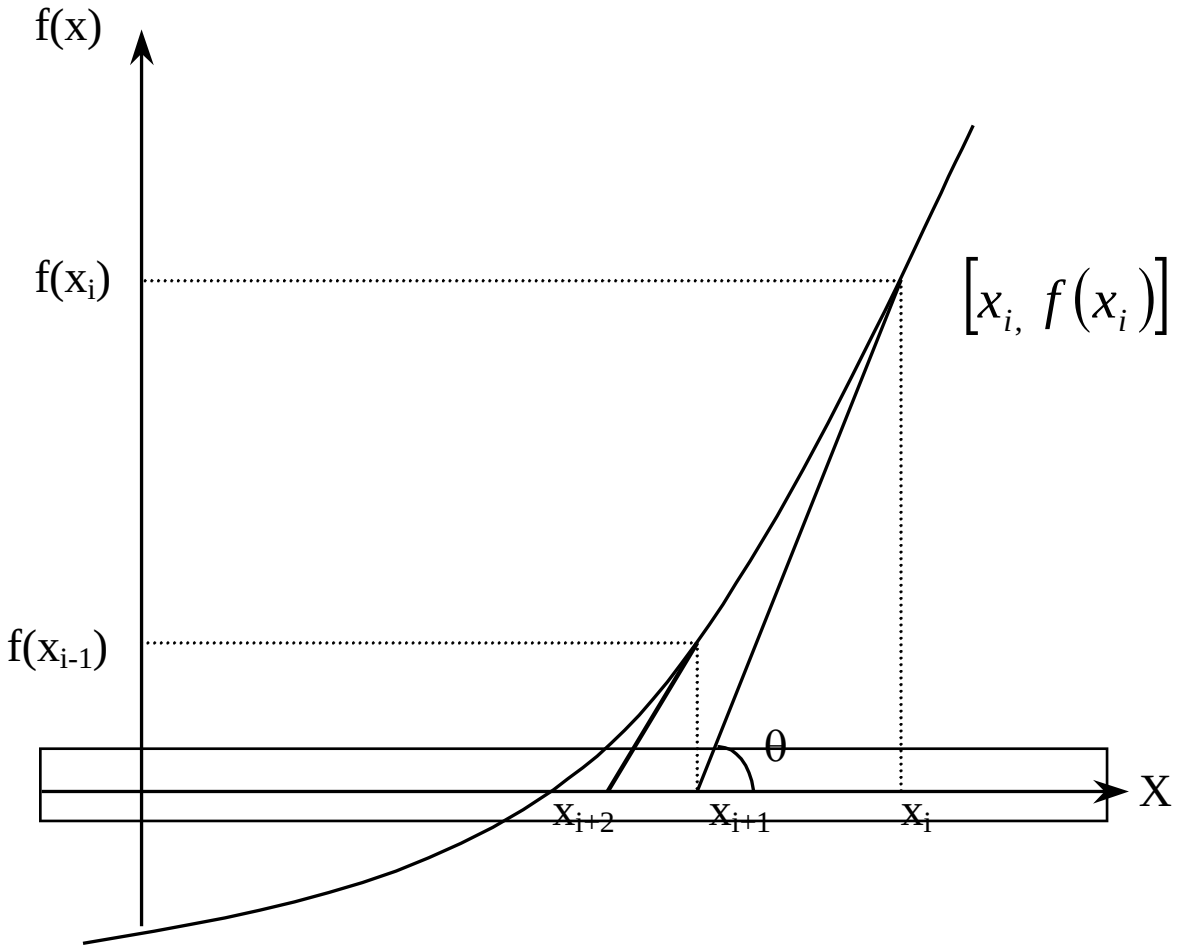
Root: Open Method

- In bracketing methods, repeated application always result in closer estimates of the true value of the root.
- Such methods are said to be convergent because they move closer to the truth as the computation progress.
- In contrast, the open methods require only a single starting value or two starting values that do not necessarily bracket the root.
- As such, they sometimes diverge or move away from the true root as the computation progresses
- However, when the open methods converge they usually faster than the bracketing methods.

Fixed Point Iteration

- A formula can be developed for simple fixed-point iteration by rearranging the function $f(x) = 0$, so that x is on the left-hand side of the equation: $x = g(x)$ (1)
- This transformation can be accomplished either by algebraic manipulation or by simply adding x to both sides of the original equation.
- The utility of Eq. (1) is that it provides a formula to predict a new value of x as a function of an old value of x.
- Thus, given an initial guess at the root x_i , Eq. (1) can be used to compute a new estimate x_{i+1} as expressed by the iterative formula $x_{i+1} = g(x_i)$ ... (2)
- As with many other iterative formulas, the approximate error for this equation can be determined using the error estimator:
- $\varepsilon_a = \left|\frac{x_{i+1} - xi}{x_{i+1}}\right| \times 100\%$ … (3)

Newton-Raphson Method

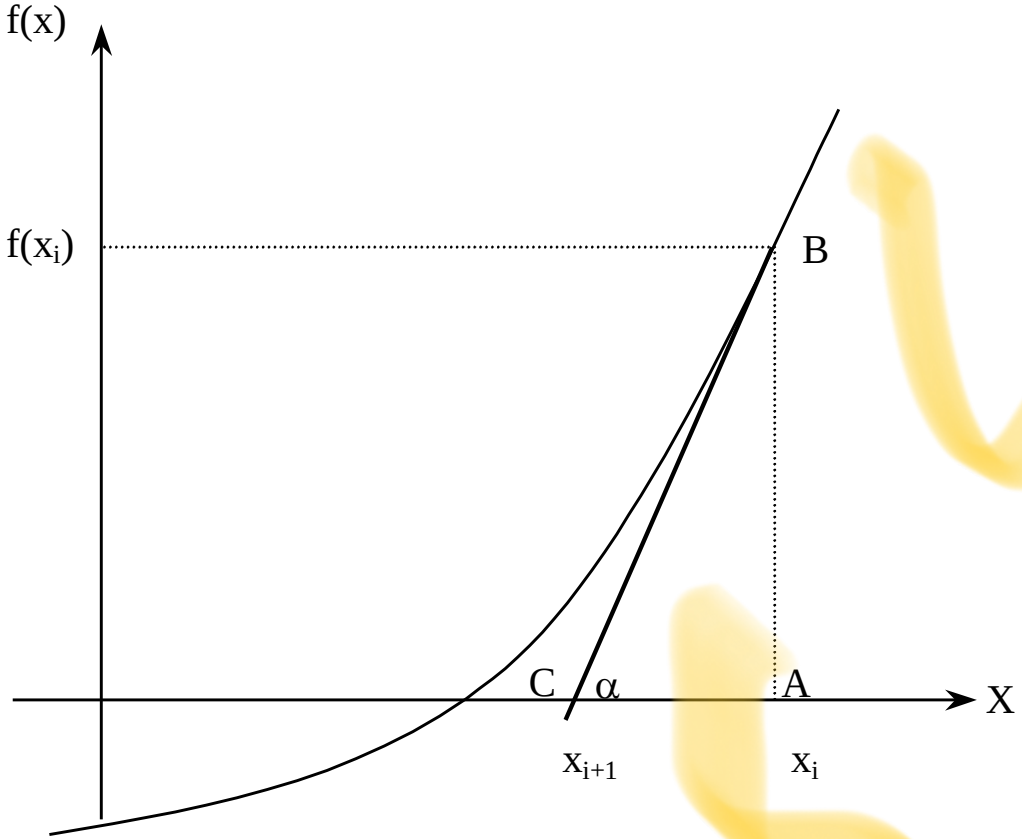


$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

$$x_{i+2} = x_{i+1} - \frac{f(x_{i+1})}{f'(x_{i+1})}$$

Figure 1 Geometrical illustration of the Newton-Raphson method.

Derivation



$$\tan(\alpha) = \frac{AB}{AC}$$

$$f'(x_i) = \frac{f(x_i)}{x_i - x_{i+1}}$$

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

Figure 2 Derivation of the Newton-Raphson method.

Step 1

<u>Example:</u> Given, $f(x) = e^{-x} - x$,

Evaluate $f'(x)$ mathematically or symbolically,

$d(e^{-x} - x) = -e^{-x} - 1$

Step 2

Use an initial guess of the root, x_i, to estimate the new value of the root, x_{i+1} , as

$$x_{i+1} = x_i - \frac{f(x_i)}{f'(x_i)}$$

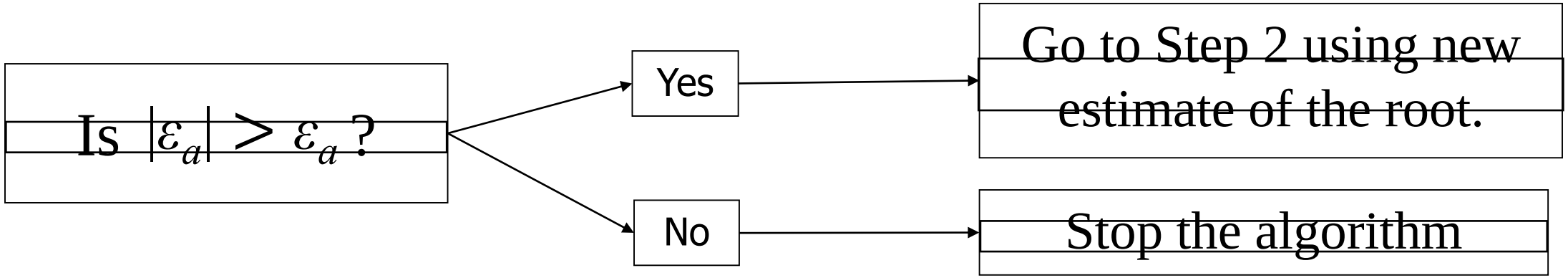
Step 3

Find the absolute relative approximate error $|\varepsilon_a|$ as;

$$|\varepsilon_a| = \left|\frac{x_{i+1} - x_i}{x_{i+1}}\right| \times 100$$

Step 4

Compare the absolute relative approximate error with the pre-specified relative error tolerance ε_s



Also, check if the number of iterations has exceeded the maximum number of iterations allowed. If so, one needs to terminate the algorithm and notify the user.

Example 1

The floating ball has a specific gravity of *0.6* and has a radius of *5.5 cm*. You are asked to find the depth to which the ball is submerged when floating in water.

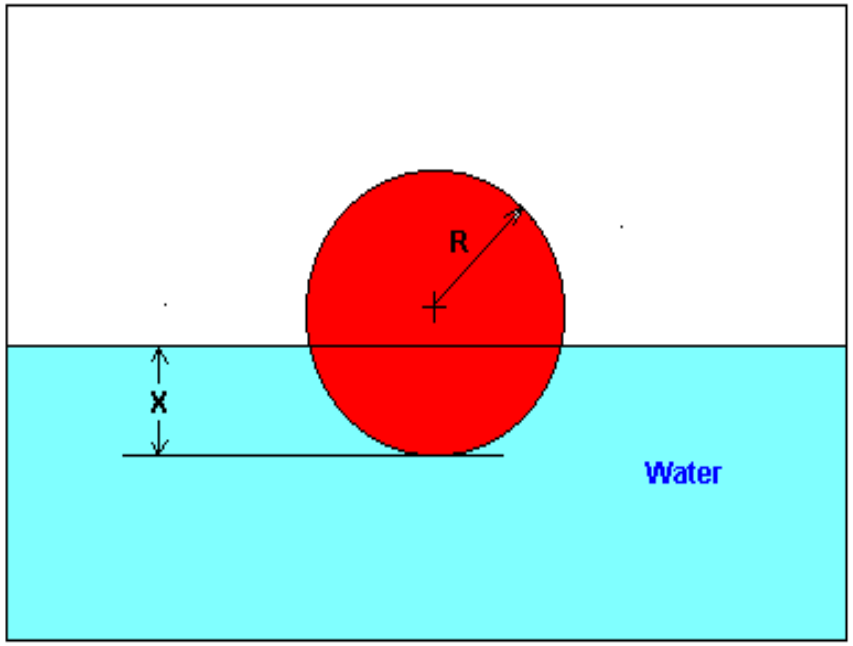


Figure 3 Floating ball problem.

Example 1 Cont.

The equation that gives the depth x in meters to which the ball is submerged under water is given by

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

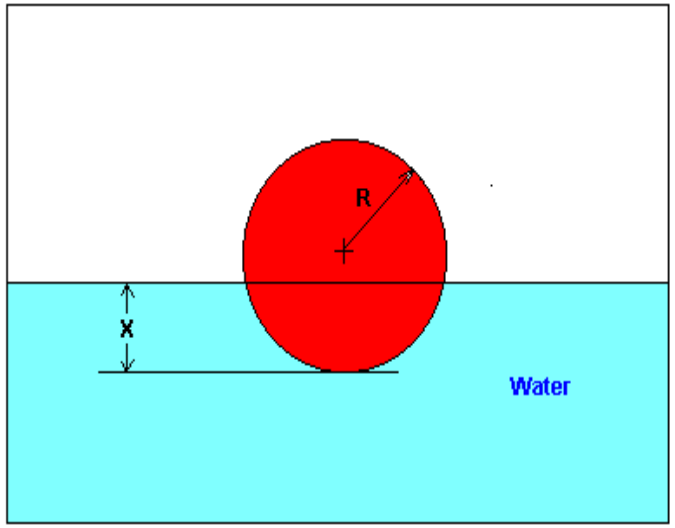


Figure 3 Floating ball problem.

Use the Newton-Raphson method of finding roots of equations to find

a) the depth 'x' to which the ball is submerged under water. Conduct three iterations to estimate the root of the above equation.

b) The absolute relative approximate error at the end of each iteration,

c) The number of significant digits at least correct at the end of each iteration.

Example 1 Cont.

Solution

To aid in the understanding of how this method works to find the root of an equation, the graph of *f(x)* is shown to the right,

where

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$

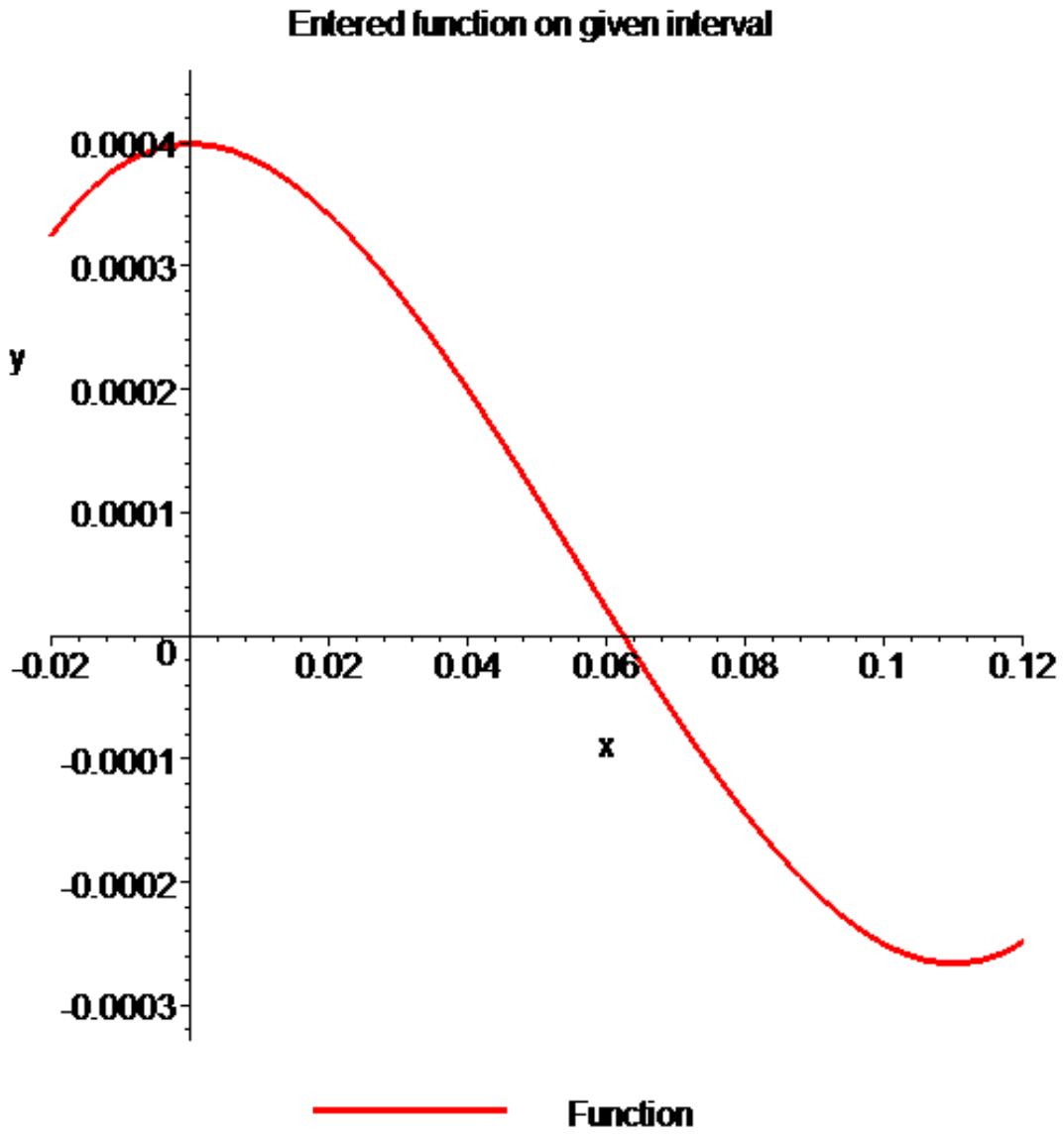


Figure 4 Graph of the function *f(x)*

Example 1 Cont.

Step-1: Evaluate $f'(x)$ mathematically or symbolically

$$f(x) = x^3 - 0.165x^2 + 3.993 \times 10^{-4}$$
$$f'(x) = 3x^2 - 0.33x$$

Step-2: Let us assume the initial guess of the root of $f(x) = 0$ for $x_0 = 0.05$m. This is a reasonable guess (discuss why $x = 0$ and $x = 0.11$m are not good choices) as the extreme values of the depth x would be 0 and the diameter (0.11 m) of the ball.

Example 1 Cont. (Step-2)

Iteration 1

The estimate of the root is

$$x_1 = x_0 - \frac{f(x_0)}{f'(x_0)}$$

$$= 0.05 - \frac{(0.05)^3 - 0.165(0.05)^2 + 3.993\times10^{-4}}{3(0.05)^2 - 0.33(0.05)}$$

$$= 0.05 - \frac{1.118\times10^{-4}}{-9\times10^{-3}}$$

$$= 0.05 - (-0.01242)$$

$$= 0.06242$$

Example 1 Cont.

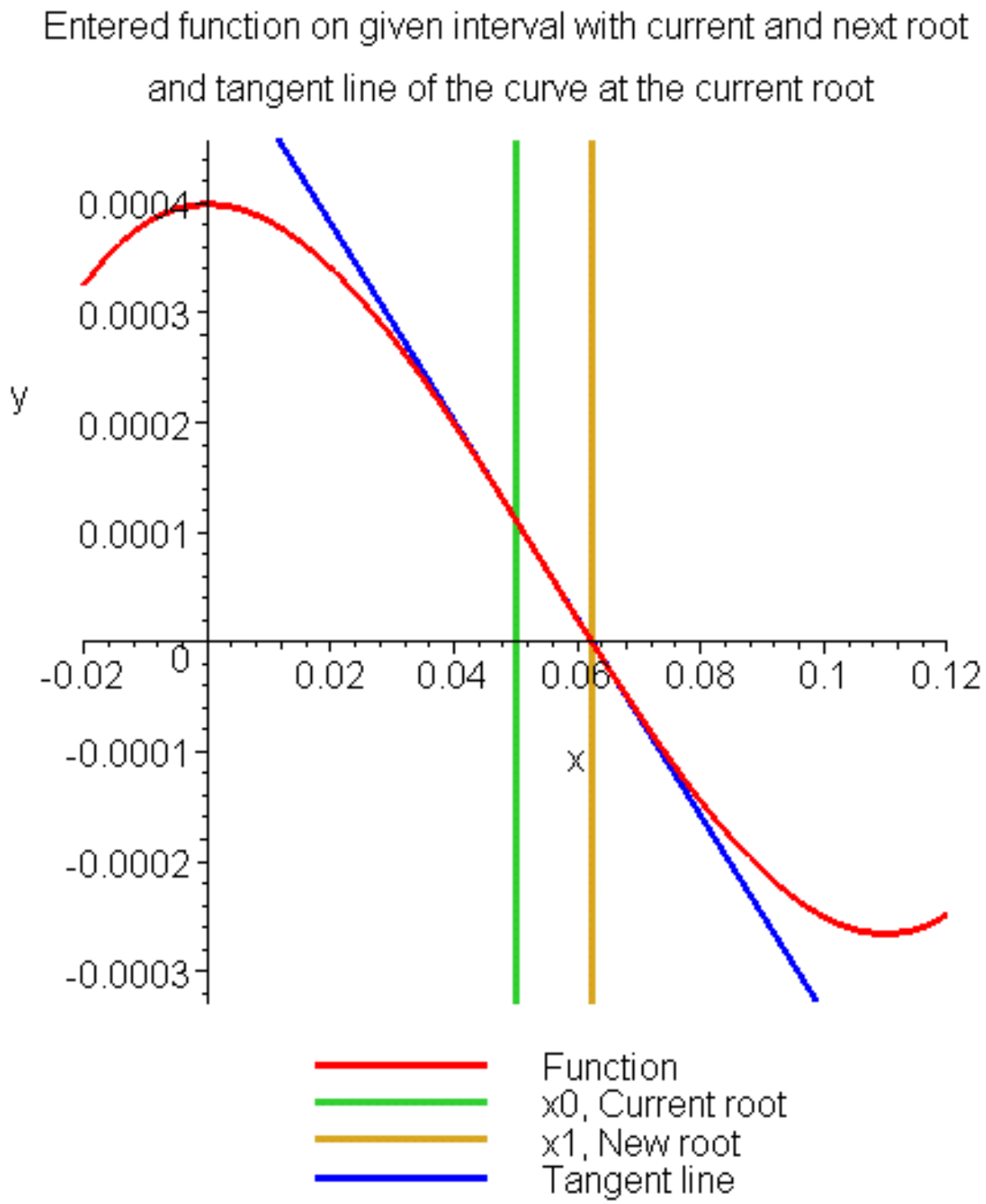


Figure 5 Estimate of the root for the first iteration.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 1 is

$$|\varepsilon_a| = \left|\frac{x_1 - x_0}{x_1}\right| \times 100$$
$$= \left|\frac{0.06242 - 0.05}{0.06242}\right|$$
$$= 19.90\%$$

The number of significant digits at least correct is 0, as you need an absolute relative approximate error of 5% or less for at least one significant digits to be correct in your result.

Example 1 Cont.

<u>Iteration 2</u>

The estimate of the root is

$$x_2 = x_1 - \frac{f(x_1)}{f'(x_1)}$$

$$= 0.06242 - \frac{(0.06242)^3 - 0.165(0.06242)^2 + 3.993 \times 10^{-4}}{3(0.06242)^2 - 0.33(0.06242)}$$

$$= 0.06242 - \frac{-3.97781 \times 10^{-7}}{-8.90973 \times 10^{-3}}$$

$$= 0.06242 - (4.4646 \times 10^{-5})$$

$$= 0.06238$$

Example 1 Cont.

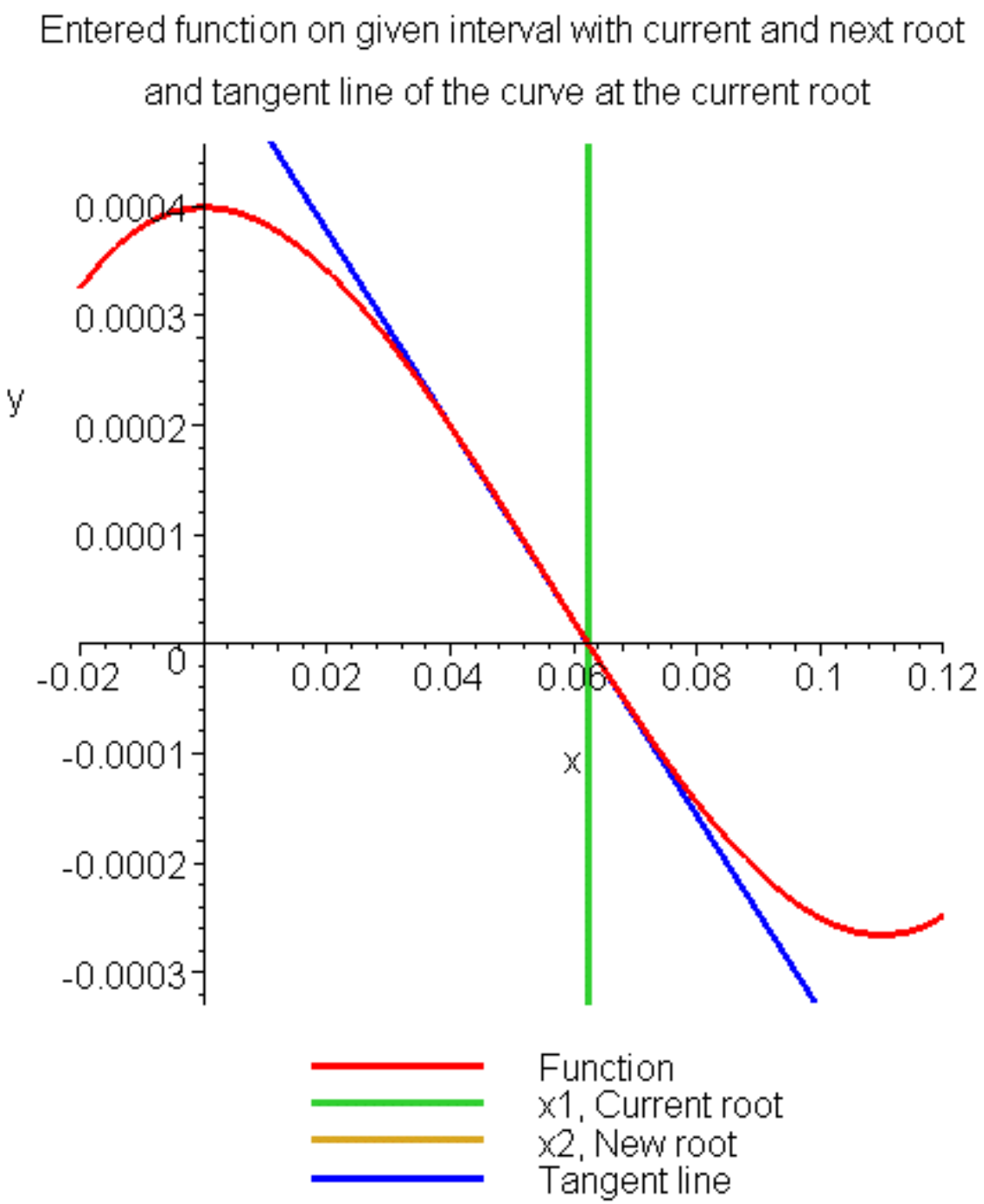


Figure 6 Estimate of the root for the Iteration 2.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 2 is

$$
\begin{aligned}
|\varepsilon_a| &= \left|\frac{x_2 - x_1}{x_2}\right| \times 100 \\
&= \left|\frac{0.06238 - 0.06242}{0.06238}\right| \\
&= 0.0716\%
\end{aligned}
$$

The maximum value of m for which $|\epsilon_a| \leq 0.5 \times 10^{2-m}$ is 2.844. Hence, the number of significant digits at least correct in the answer is 2.

Example 1 Cont.

<u>Iteration 3</u>

The estimate of the root is

$$x_3 = x_2 - \frac{f(x_2)}{f'(x_2)}$$

$$= 0.06238 - \frac{(0.06238)^3 - 0.165(0.06238)^2 + 3.993\times10^{-4}}{3(0.06238)^2 - 0.33(0.06238)}$$

$$= 0.06238 - \frac{4.44\times10^{-11}}{-8.91171\times10^{-3}}$$

$$= 0.06238 - \left(-4.9822\times10^{-9}\right)$$

$$= 0.06238$$

Example 1 Cont.

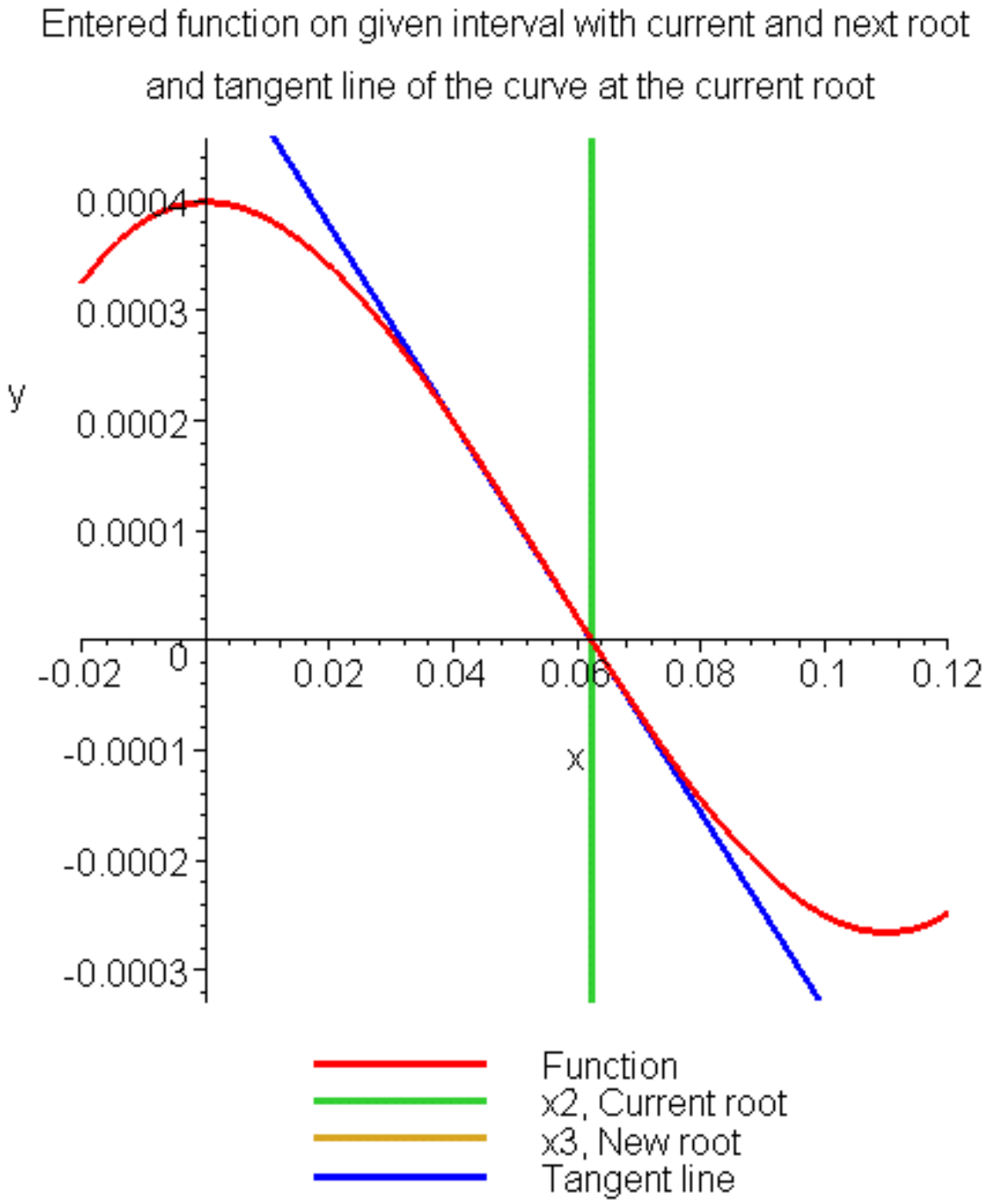


Figure 7 Estimate of the root for the Iteration 3.

Example 1 Cont.

The absolute relative approximate error $|\varepsilon_a|$ at the end of Iteration 3 is

$$
\begin{aligned}
|\varepsilon_a| &= \left|\frac{x_2 - x_1}{x_2}\right| \times 100 \\
&= \left|\frac{0.06238 - 0.06238}{0.06238}\right| \times 100 \\
&= 0\%
\end{aligned}
$$

The number of significant digits at least correct is 4, as only 4 significant digits are carried through all the calculations.

Conclusion

- As with other root-location methods, Eq. (3) can be used as a termination criterion.
- In addition, a theoretical analysis (Chapra and Canale, 2010) provides insight regarding the rate of convergence as expressed by

$$\varepsilon_{t,i+1} = -\frac{f''(x_r)}{2f'(x_r)}\varepsilon_{t,i}^2$$

- Thus, the error should be roughly proportional to the square of the previous error. In other words, the number of significant figures of accuracy approximately doubles with each iteration. This behavior is called quadratic convergence and is one of the major reasons for the popularity of the method.